Rare Disease, Real Kid: The MVA Hackathon 2026

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Model number: 1

Abstract

We propose five drug-repurposing candidates for an MVA patient carrying two predicted loss-of-function variants in *BUB1B*. The approved proteasome inhibitor bortezomib ranks first, followed by WEE1 inhibitors and KIF18A inhibitors that are on research. This were selected as each target a vulnerability that the literature reports for aneuploid or chromosomally unstable cells: proteotoxic stress, replication stress with DNA damage, and dependence on the mitotic machinery.

We used the track 1 variant annotations to define the disease mechanism (biallelic loss of the spindle-checkpoint kinase BUB1B). We then reviewed the literature on targeting aneuploidy and chromosomal instability, and queried three drug–target resources (OptimusKG, DGIdb, Open Targets).

Candidates were ranked by target specificity and approval stage. All candidates are hypotheses that require further investigation. One of the limitations of this study is that all proposed drugs were developed for cancer and do not consider that BUB1B deficiency is present in all patient cells. Nonetheless this drugs may also be considered for treatment of tumors related to MVA as they will likely have high levels of aneuploidy and chromosomal instability.

Approach description

Rationale

As restoring BUB1B function is no possible with existing treatments, we looked to identify specific characteristics of aneuploid/chromosomally unstable cells that could be targeted so normal diploid cells remain unaffected. We then searched for existing drugs that act on those properties.

Identification of candidates for drug repurposing

To identify any existing and approved medications that could potentially benefit MVA patients with the same or similar variants than the ones that were identified in track one we first used the consequence prediction annotation previously obtained from track one to establish the variant mechanism. Results from track one showed that two heterozygous compound variants were affecting the gene BUB1B, this variants were predicted as a stop gain and a missense variant. A stop gain is a high impact variant that can truncate a protein affecting its ability to work properly. This variant was already

annotated as pathogenic or likely pathogenic in the ClinVar database. The effect of a missense variant will depend on the position and the change of amino acid that it generates. To predict this effect there are several tools that estimate its deleteriousness based on sequence homology, conservation between species, and through the prediction of physical and chemical changes that may be generated in the protein by the amino acid change. Variants were also annotated with this type of tools on track one. For this missense variant the prediction was that it was deleterious and probably damaging. Therefore we expect that this two variants are causing a balletic inactivation of the gene.

We researched more thoroughly the function of the gene affected on the NCBI gene database. This gene is a kinase that is involved in spindle checkpoint function, delaying the onset of anaphase and ensuring proper chromosome segregation. This was expected as it is a process that is affected in MVA as reported in the literature. Through literature research both about loss of function of BUB1B and MVA we identified that in both cases there was evidence of the presence of chromosomal instability, which just as aneuploidy is commonly observed in several types of cancer. Chromosomal instability has also been researched as a potential target for cancer treatment.

Considering this data we used both literature research regarding targeting chromosomal instability or aneuploidy and drug - gene interaction databases query focusing on genes/ pathways mentioned in literature as potential vulnerabilities for chromosomal instability/ aneuploidy. After identifying potential drug candidates both through literature research and database query we ranked the drug candidates based on the specific drug target, the pathways involved and approval or research stage.

Based on literature research we identified the following processes/characteristics of aneuploid cells that could be vulnerabilities and potential targets:

Genome and proteome imbalance in aneuploid cells that the cell overcomes by by reducing protein translation and increasing protein degradation. This makes aneuploid cells more sensitive to proteasome inhibition (Ippolito et al., 2024; Al-Rawi et al., 2024).

Aneuploid cells have lower proliferative ability due to the effects of proteotoxic stress and induction of DNA damage. Previous studies have shown that inhibiting WEE1 kinase increases the levels of replication stress and DNA damage in the cell and subsequently cause severe chromosome pulverization and therefore causing cell death (Vriend et al., 2013; Haykal et al., 2023; Al-Rawi et al., 2024).

Aneuploid cells have abnormal mitotic phenotypes more frequently higher mitotic times and increased errors, and therefore are more dependent on mitotic machinery for a successful mitosis. Previous studies have shown that loss of KIF18A in aneuploid or chromosomally unstable cells results in mitotic defects and eventual cell death(Al-Rawi et al., 2024; Phillips et al., 2025).

Ranked candidates

- bortezomib (CHEMBL325041) - Proteasome inhibitor. Approved.
- fostamatinib (CHEMBL2103830) - Off targets WEE1. Suppresses the same downstream cell-cycle checkpoint that WEE1 regulates. Approved.
- azenosertinb (CHEMBL5095036) - WEE1 inhibitor. Not approved but on clinical trials.
- adavosertib (CHEMBL1976040) - WEE1 inhibitor. Not approved but on clinical trials.
- KIF18A inhibitors - On research.

Candidate identification: automated or manual

This approach used manual literature review as well as the creation and use of scripts to query drug-target databases using as input a gene or drug name.

Manual literature review

Manual literature review was done using search engines as google as well as the NCBI resources looking for papers related to MVA and targeting chromosomal instability. Both, open access papers and subscription access papers were used.

Publicly available databases and literature

For this analysis we used publicly available drug databases

Databases used:

Optimuskg - a multimodal biomedical labeled property graph intended for biomedical discovery
Dgidb - drug - gene interaction database that comprises drug and gene data from publications and expert resources
Opentargets - tool for identification and prioritization of potential therapeutic drug targets. Integrates several public datasets.

For literature review both open access and subscription access papers were used.

Variant mechanism characterization

Variant mechanism characterization was done based on the annotations that were generated through track 1, which included the annotation of the type of variant (stop gain, missense, frameshift insertion, etc.), the prediction of the consequence of missense variants through the tools SIFT and PolyPhen. ClinVar pathogenicity annotation was also considered. And further literature review and data query was done to determine biological consequence and pathways involved.

Estimated time spent on this analysis

We dedicated an estimated 72 hours between literature review, script generation, databases query and candidate ranking.

Generative AI used

Anthropic, Claude Opus, commercial terms, no training on customer content. Used to help write scripts and identify available drug-target databases.

References:

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